Inner ear & IAC

Bony and membranous labyrinth, the fluids, organ of Corti, vestibular receptors, blood supply and the fundus of the IAC.

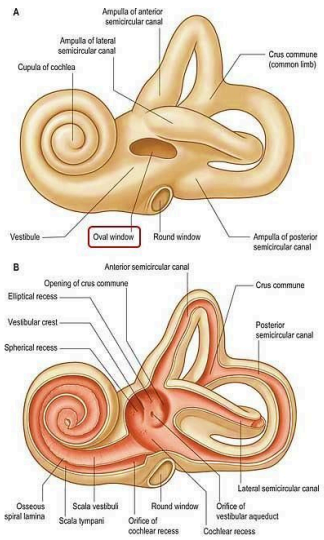
1. Bony labyrinth
2. Scalae and windows
3. Membranous labyrinth
4. Inner ear fluids
5. Organ of Corti
6. Vestibular receptors
7. Blood supply of the labyrinth
8. Internal auditory canal
9. Central auditory and vestibular pathways
10. On imaging
11. In theatre

01 Bony labyrinth

- **Vestibule** (central chamber, 5 mm): **oval window** in the lateral wall; medial wall has the **spherical recess** (saccule, in front) and **elliptical recess** (utricle, behind), separated by the vestibular crest; opening of the **vestibular aqueduct** below the elliptical recess.
- **Semicircular canals:** each two-thirds of a circle, 0.8 mm diameter, at right angles to each other. They open into the vestibule by **5 openings**: three ampullated ends, the non-ampullated end of the lateral canal, and the **crus commune** of the superior and posterior canals.
- **Lateral canal** tilts 30° from the horizontal and bulges into the aditus and antrum. **Superior canal** lies across the petrous axis (arcuate eminence). **Posterior canal** parallels the posterior surface of the petrous bone.
- **Cochlea: 2.5–2.75 turns** around the **modiolus**; about 30–35 mm long, 5 mm base to apex. The **osseous spiral lamina** winds round the modiolus and carries the basilar membrane; **Rosenthal's canal** holds the spiral ganglion.

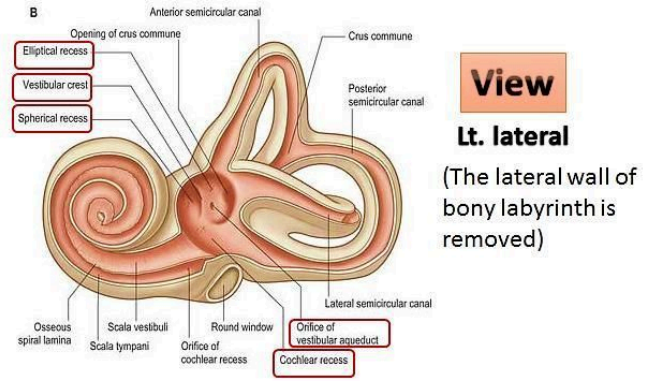
CLINICAL PEARL The modiolus base faces the IAC; a wide IAC with an absent or deficient modiolus (IP-III, X-linked; also IP-I) predisposes to CSF gusher at cochlear implantation; IP-II (Mondini) usually gives only an ooze.

SOURCES DIFFER COCHLEAR TURNS: Doss 2.5; ENT Steps 2.5–2.75.



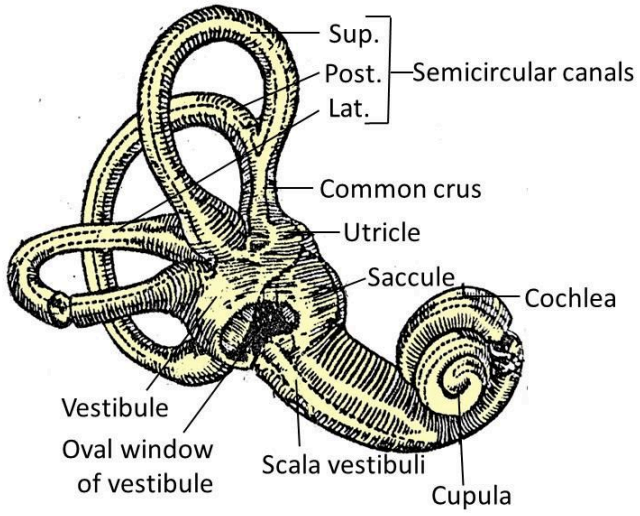
Bony labyrinth: lateral and medial walls

ENT Steps Anatomy, p. 64



Bony labyrinth with the lateral wall removed

ENT Steps Anatomy, p. 66



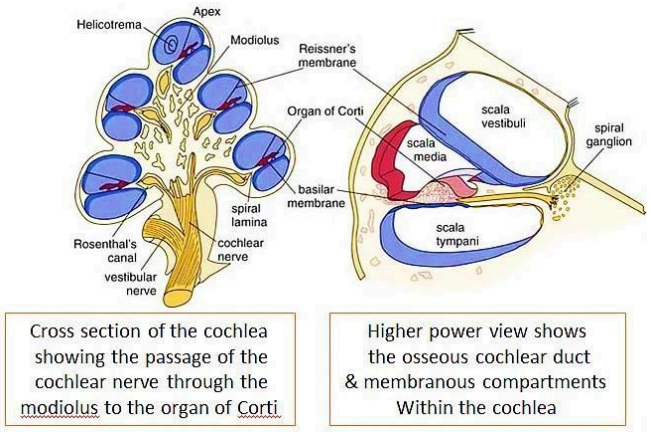
Bony labyrinth

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02 Scalae and windows

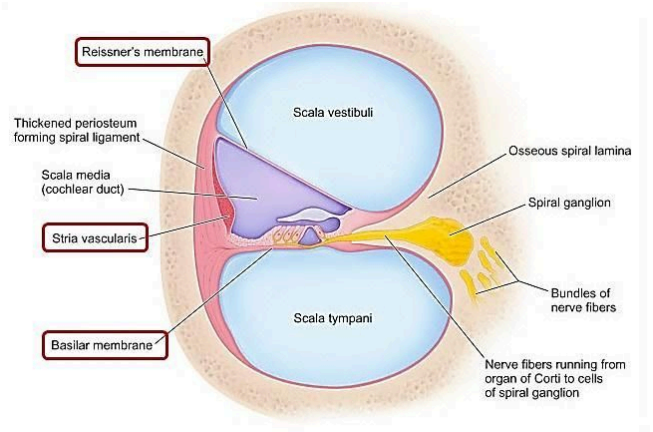
- **Scala vestibuli** (top): continuous with the vestibule, closed by the stapes footplate at the oval window.
- **Scala tympani** (bottom): closed by the **secondary TM at the round window**.
- **Scala media (cochlear duct):** triangular; floor = **basilar membrane**, roof = **Reissner's membrane**, outer wall = **stria vascularis**.
- **Helicotrema** joins scala vestibuli and tympani at the apex.
- **Cochlear aqueduct** connects scala tympani to the subarachnoid space.

> **CLINICAL PEARL** Cochlear implant electrodes go into the scala tympani, through the round window or a cochleostomy just anteroinferior to it.



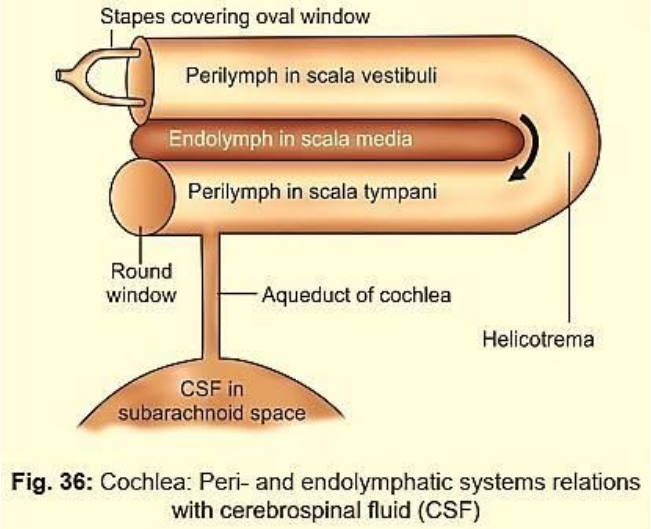
Cross-section of the cochlea and modiolus

ENT Steps Anatomy, p. 69



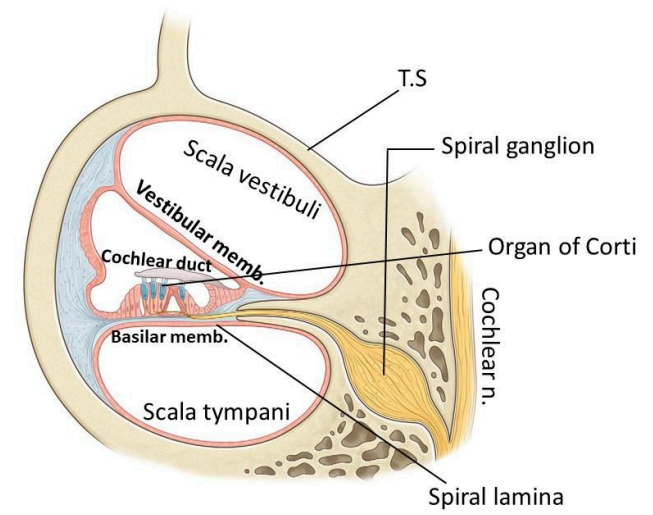
Cochlear duct in cross-section

ENT Steps Anatomy, p. 72



Peri- and endolymphatic spaces of the cochlea and the cochlear aqueduct

ENT Steps Anatomy, p. 75



Cochlear duct in cross-section

Dr. Sameh Doss handout, p. 210

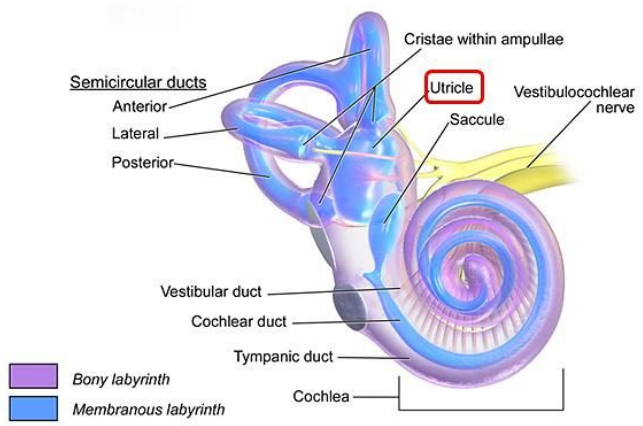
03 Membranous labyrinth

- Cochlear duct, utricle, saccule, three semicircular ducts, endolymphatic duct and sac.
- **Basilar membrane widens from base to apex:** high frequencies at the base, low at the apex (tonotopy).
- **Utricle** (posterior vestibule) receives the 5 openings of the semicircular ducts; **saccule** lies in front, opposite the stapes footplate, and joins the cochlear duct by the **ductus reuniens**.
- The **endolymphatic duct** runs in the vestibular aqueduct to the **endolymphatic sac**, between two layers of dura on the posterior petrous surface. It absorbs and regulates endolymph.

> **CLINICAL PEARL** The saccule sits right under the footplate, so it is the structure at risk from an overly deep stapedotomy piston.

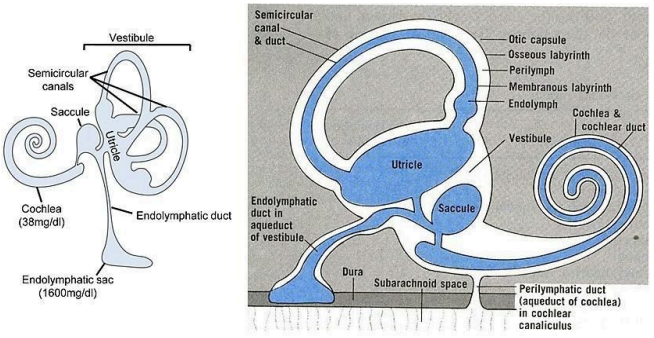


Tonotopy. High-frequency hair cells at the base meet every sound first, which is why presbycusis, ototoxicity and noise damage start in the high frequencies.



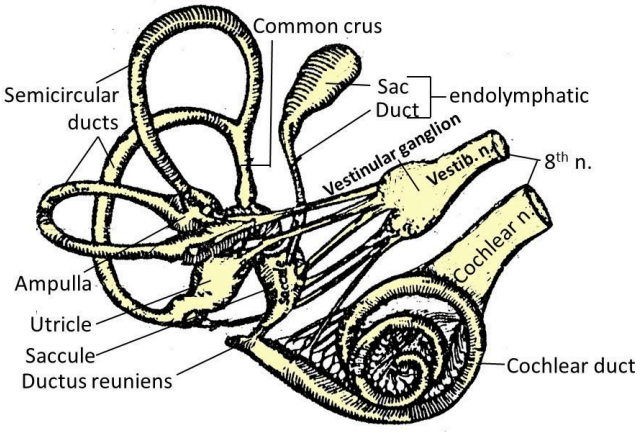
Membranous labyrinth within the bony labyrinth

ENT Steps Anatomy, p. 72



Membranous labyrinth and endolymphatic sac

ENT Steps Anatomy, p. 74



Membranous labyrinth

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04 Inner ear fluids

- **Perilymph** (between bony and membranous labyrinth): like extracellular fluid, **high Na+**, low K+. From blood filtrate in the spiral ligament and from CSF via the cochlear aqueduct.
- **Endolymph** (inside the membranous labyrinth): like intracellular fluid, **high K+ (~150 mEq/L)**, very low Na+. Secreted by the **stria vascularis** and dark cells of the vestibule.
- Absorption: **longitudinal flow** to the endolymphatic sac, and **radial flow** with secretion and absorption at the stria.

CLINICAL PEARL Ménière's disease = endolymphatic hydrops; the sac is the target of endolymphatic sac surgery.

TABLE 2 Differences between the composition of endolymph, perilymph and CSF**

	Endolymph	Perilymph	CSF
Na^+ (mEq/L)	3	150	152
K^+ (mEq/L)	150	3–5	4
Cl^- (mEq/L)	130	125	
Protein (mg/dL)	126	200–400	20–50
Glucose (mg/dL)	10–40	85	70

**Values vary from the site of collection such as cochlea, saccule and endolymphatic sac in cases of endolymph and scala tympani and vestibuli in cases of perilymph.

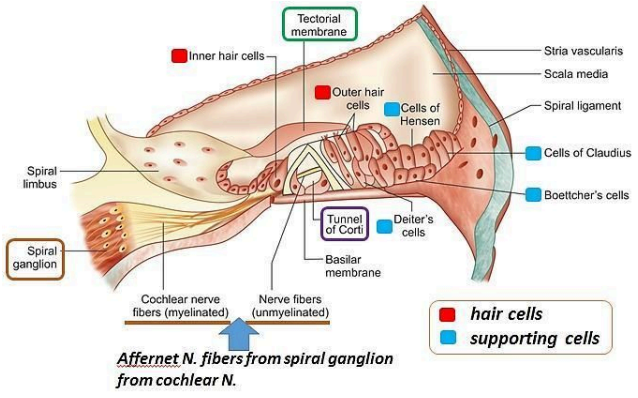
Composition of endolymph, perilymph and CSF

ENT Steps Anatomy, p. 75

05 Organ of Corti

- Sits on the basilar membrane along its whole length. Tunnel of Corti between inner and outer rods contains cortilymph.
- **Inner hair cells:** about 3,500, one row, flask-shaped, receive **95% of afferent** spiral ganglion fibres; they transmit sound.
- **Outer hair cells:** about 12,000, 3–4 rows, cylindrical, mainly **efferent** from the superior olivary complex; they amplify and tune (cochlear amplifier) and generate **otoacoustic emissions**.
- OHCs are **more sensitive to ototoxic drugs and noise** than IHCs.
- Supporting cells: Deiters' (between OHCs), Hensen's. The **tectorial membrane** lies over the hair cells; shearing between them bends the stereocilia.

CLINICAL PEARL Absent OAEs with present ABR waves points to outer hair cell loss; present OAEs with abnormal ABR points to auditory neuropathy.



Organ of Corti

ENT Steps Anatomy, p. 77

06 Vestibular receptors

- **Cristae** in the ampullae detect **angular acceleration**. Hair cells project into the gelatinous **cupula**, which spans the ampulla like a swinging door.

- **Maculae** of the utricle (horizontal, larger, ~33,000 hair cells) and saccule (vertical, ~18,000) detect **linear acceleration and gravity**. They carry an otolithic membrane with **otoconia** (calcium carbonate).
- Each hair cell has many stereocilia and **one kinocilium**. Type I cells are flask-shaped with a calyx ending; type II are cylindrical with bouton endings.
- The **striola** divides each macula; polarity reverses across it.
- In the **lateral canal**, endolymph flow **toward the utricle (ampullopetal) excites**; in the superior and posterior canals, flow **away from the utricle excites** (Ewald).

> CLINICAL PEARL Otoconia displaced from the utricle into the posterior canal cause posterior canal BPPV, the commonest peripheral vertigo.

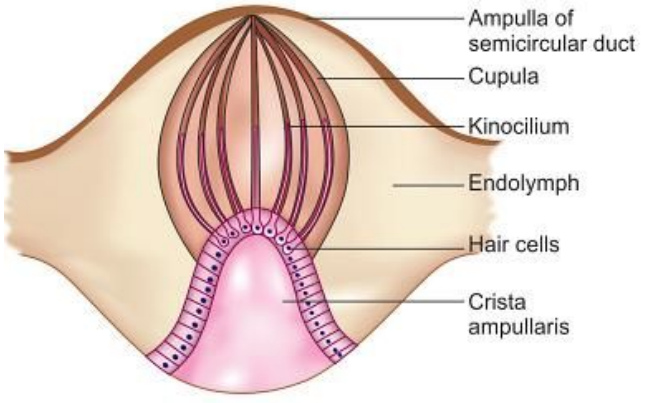


Fig. 41: Crista, hair cells and cupula. Cut section of ampulla of semicircular duct

Crista ampullaris and cupula

ENT Steps Anatomy, p. 78

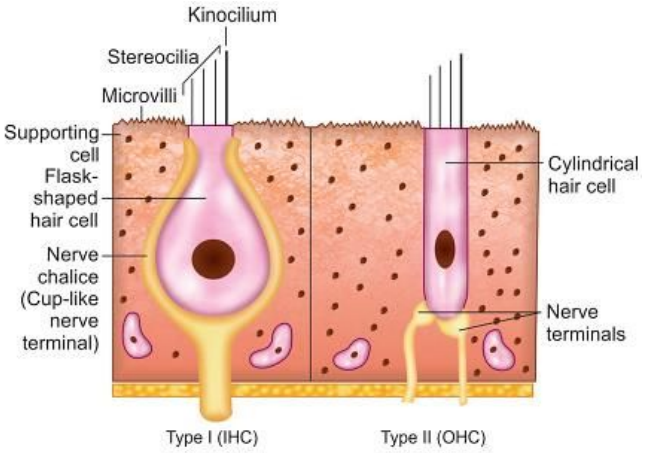


Fig. 42: Hair cells of vestibular organs

Type I and type II vestibular hair cells

ENT Steps Anatomy, p. 79

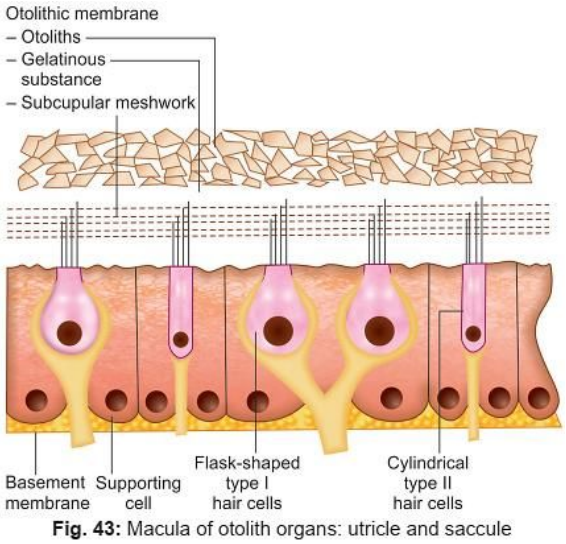


Fig. 43: Macula of otolith organs: utricle and saccule

Macula of the utricle and saccule

ENT Steps Anatomy, p. 80

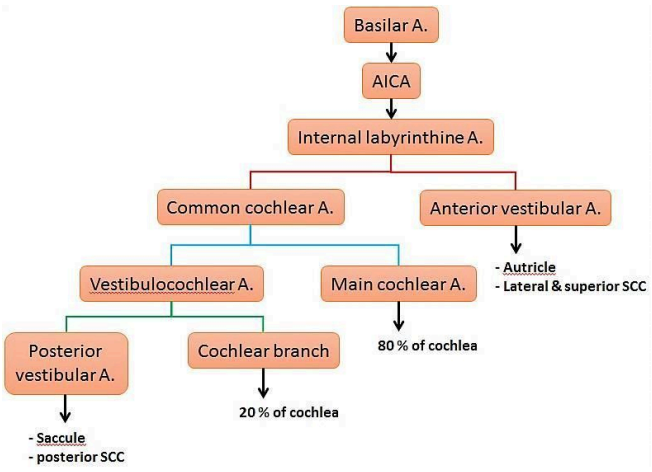
07 Blood supply of the labyrinth

- **Labyrinthine (internal auditory) artery**, usually from **AICA**, sometimes the basilar artery. It is an **end artery**.
- Divides into the anterior vestibular artery and the common cochlear artery (main cochlear and vestibulocochlear branches).

- Veins: internal auditory vein and veins of the cochlear and vestibular aqueducts, to the inferior petrosal and sigmoid sinuses.

CLINICAL PEARL End-artery supply explains sudden sensorineural hearing loss and vertigo from AICA territory strokes.

SOURCES DIFFER ORIGIN AND VEINS: ENT Steps: mostly from AICA, veins to the inferior petrosal and sigmoid sinuses. Doss: labyrinthine branch of the basilar plus the stylomastoid artery, veins to the superior petrosal or transverse sinus.



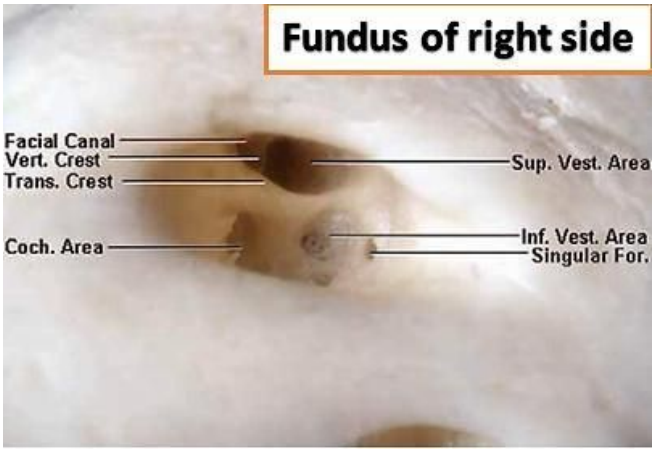
Branches of the labyrinthine artery

ENT Steps Anatomy, p. 82

08 Internal auditory canal

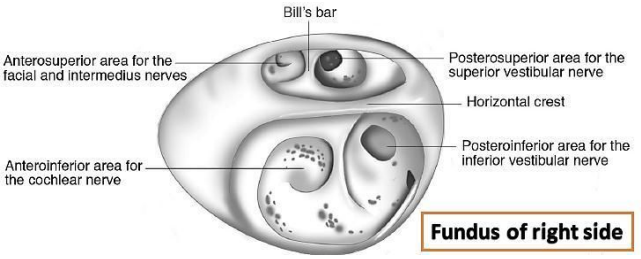
- About **1 cm long**, lined by dura, running laterally into the petrous bone. The **fundus** is closed by a cribriform plate.
- The **transverse (falciform) crest** splits the fundus into upper and lower parts; **Bill's bar** (vertical crest) splits the upper part.
- **Quadrants:** facial nerve **anterosuperior**; superior vestibular **posterosuperior**; cochlear **anteroinferior**; inferior vestibular **posteroinferior** (with the singular nerve to the posterior canal).
- **Superior vestibular nerve:** superior and lateral canals, utricle, part of the saccule. **Inferior vestibular nerve:** posterior canal and most of the saccule.
- An AICA loop can run between VII and VIII and is a landmark in vestibular schwannoma surgery.

CLINICAL PEARL Mnemonic: "Seven-Up, Coke Down" (facial above, cochlear below, both in front). Bill's bar separates the facial from the superior vestibular nerve in translabyrinthine surgery.



Fundus of the right IAC

ENT Steps Anatomy, p. 83

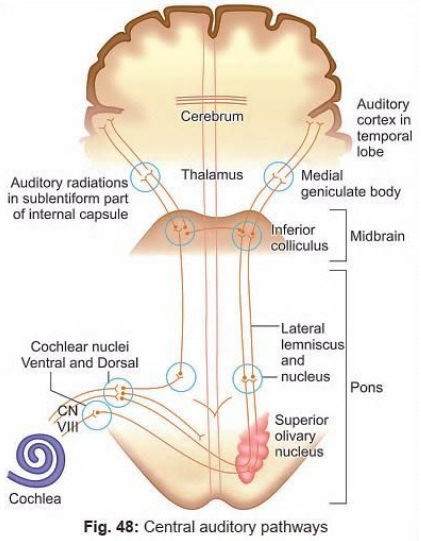


Fundus of the IAC: Bill's bar and the transverse crest

ENT Steps Anatomy, p. 83

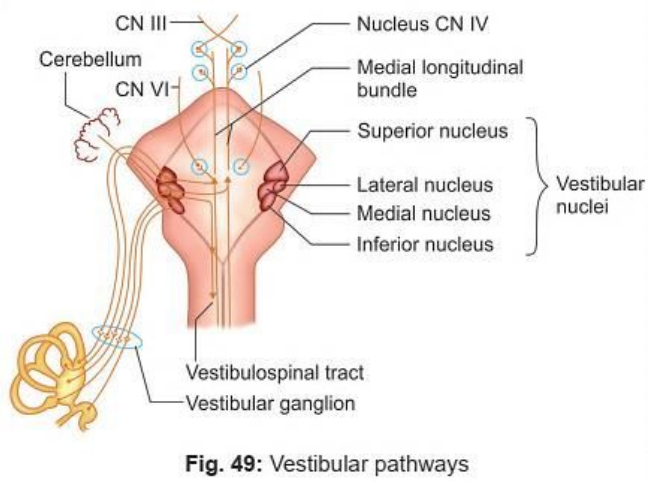
09 Central auditory and vestibular pathways

- **Auditory pathway (E-COLI-MA):** Eighth nerve → Cochlear nuclei → superior Olivary complex → Lateral lemniscus → Inferior colliculus → Medial geniculate body → Auditory cortex (superior temporal gyrus, area 41).
- Each ear projects to **both** hemispheres, so a unilateral cortical lesion does not cause unilateral deafness.
- Four vestibular nuclei: superior, inferior, medial, lateral. Outputs: **medial longitudinal fasciculus** (vestibulo-ocular reflex, nystagmus), vestibulospinal tract (posture), vestibulocerebellar tract, autonomic centres (nausea, sweating) and the temporal cortex (awareness of motion).
- Vestibular neurons fire at rest at about 90 spikes per second; asymmetry between the two sides produces vertigo.



Central auditory pathway

ENT Steps Anatomy, p. 84



Vestibular pathways

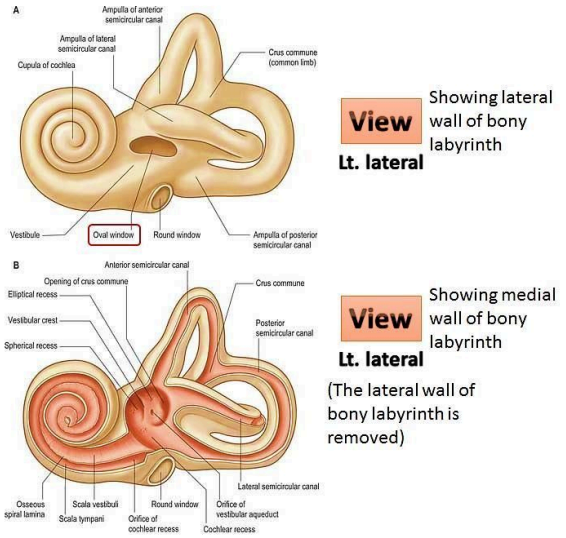
ENT Steps Anatomy, p. 86

10 Radiological anatomy: On imaging

- **CT:** bony labyrinth malformations (Mondini/IP-II, common cavity), enlarged vestibular aqueduct (over 1.5 mm at the midpoint), otosclerosis (lucency in front of the oval window: fissula ante fenestram), cochlear ossification after meningitis.

- **MRI (T2/CISS):** fluid signal in the cochlea (patency before implantation), cochlear nerve size in the IAC, vestibular schwannoma (enhancing on T1 with gadolinium).
- Cochlear ossification after meningitis can start within weeks, so implant early.

> **CLINICAL PEARL** Enlarged vestibular aqueduct is the commonest inner ear anomaly on imaging in children with sensorineural loss; avoid head trauma and contact sports.



Bony labyrinth: lateral and medial walls
ENT Steps Anatomy, p. 64

11 Surgical anatomy: In theatre

- **Cochlear implantation:** mastoidectomy → posterior tympanotomy → round window or cochleostomy anteroinferior to it → electrode into the scala tympani.
- **Translabyrinthine approach:** removes the labyrinth to reach the IAC and CPA; hearing is sacrificed; Bill's bar identifies the facial nerve at the fundus.
- **Labyrinthectomy and vestibular neurectomy** for intractable vertigo; the singular nerve supplies the posterior canal (singular neurectomy for BPPV, rarely done).

T Tables

Endolymph vs perilymph

	Endolymph	Perilymph
Location	Inside the membranous labyrinth	Between bony and membranous labyrinth
Resembles	Intracellular fluid	Extracellular fluid / CSF
K+	High (~150 mEq/L)	Low (3–5 mEq/L)
Na+	Low (~3 mEq/L)	High (~150 mEq/L)
Source	Stria vascularis, vestibular dark cells	Blood filtrate (spiral ligament), CSF via cochlear aqueduct
Absorption	Endolymphatic sac (longitudinal flow), stria (radial flow)	Venous and CSF routes

Ménière's disease = endolymphatic hydrops.

Inner vs outer hair cells

	Inner hair cells	Outer hair cells
Number	~3,500	~12,000
Rows	1	3–4
Shape	Flask	Cylindrical
Innervation	95% of afferent fibres	Mainly efferent (olivocochlear)
Role	Transduce sound to the brain	Amplify and tune (cochlear amplifier)
Otoacoustic emissions	No	Yes
Ototoxicity and noise	More resistant	Damaged first

Vestibular nerve branches

Branch	Supplies
Superior vestibular nerve	Cristae of superior and lateral canals, utricle, anterosuperior saccule
Inferior vestibular nerve	Crista of posterior canal (singular nerve), most of the saccule

Caloric test checks the lateral canal = superior vestibular nerve. cVEMP checks the saccule = inferior vestibular nerve.

Tuning fork tests

Finding	Rinne (affected ear)	Weber
Normal	Positive (AC > BC)	Central
Conductive loss	Negative (BC > AC) when the air-bone gap is about 25–30 dB or more with 512 Hz (15–20 dB with 256 Hz)	Lateralises to the affected ear
Sensorineural loss	Positive (both reduced)	Lateralises to the better ear
Dead ear	False negative (sound heard in the other ear by bone)	To the better ear

Use a 512 Hz fork.

Development of the ear

Part	Origin	Timing / notes
Inner ear	Otic placode → otic vesicle	Week 4; adult size by ~week 20–25
Middle ear and ET	1st pouch (tubotympanic recess)	Pneumatisation continues after birth
Malleus head, incus body	1st arch	—
Handle of malleus, long process of incus, stapes superstructure	2nd arch	Footplate partly otic capsule
EAC	1st cleft (meatal plug canalises)	Weeks 21–28
Auricle	Six hillocks of His (1st and 2nd arches)	Weeks 6–8
Mastoid process	Grows after birth	Absent at birth; facial nerve superficial

M Mnemonics

IAC nerve positions

Seven-Up, Coke Down

Facial nerve anterosuperior; cochlear nerve anteroinferior

Auditory pathway

E-COLI-MA

Eighth nerve, Cochlear nuclei, superior Olive, Lateral lemniscus, Inferior colliculus, Medial geniculate, Auditory cortex

Inner ear fluids

PotassIUM is INSIDE (endolymph)

Endolymph is potassium-rich like intracellular fluid

Ions

Endolymph = K, Perilymph = Na

Potassium inside the membranous labyrinth, sodium outside

Canal openings

Three canals, five holes

Three ampullae, one lateral non-ampullated end, one crus commune

Tonotopy

High at the hook

High frequencies at the basal (hook) end, low at the apex

Q Rapid recall

1. **How many turns does the cochlea make?** 2.5 to 2.75 turns around the modiolus.
2. **How many openings do the semicircular canals have into the vestibule?** Five: three ampullated ends, the non-ampullated lateral canal end and the crus commune.
3. **Ionic composition of endolymph and perilymph?** Endolymph: high K+ (~150), low Na+. Perilymph: high Na+, low K+.
4. **Which structure secretes endolymph?** Stria vascularis (and dark cells of the vestibule).
5. **Inner vs outer hair cells: innervation and number?** IHC ~3,500 in one row, 95% of afferents. OHC ~12,000 in 3–4 rows, mainly efferent; generate OAEs.
6. **Where are high frequencies coded on the basilar membrane?** At the base (narrow, stiff); low frequencies at the apex.
7. **What do the cristae and maculae detect?** Cristae: angular acceleration. Maculae of utricle and saccule: linear acceleration and gravity.
8. **Arrangement of nerves at the IAC fundus?** Facial anterosuperior, superior vestibular posterosuperior, cochlear anteroinferior, inferior vestibular posteroinferior. Bill's bar separates the upper two.
9. **Blood supply of the labyrinth?** Labyrinthine artery, usually from AICA; an end artery.
10. **List the central auditory pathway.** Cochlear nerve → cochlear nuclei → superior olivary complex → lateral lemniscus → inferior colliculus → medial geniculate body → auditory cortex (area 41).
11. **Diameter of each semicircular canal?** About 0.8 mm; each forms about two-thirds of a circle.
12. **Which canal bulges into the aditus and antrum?** The lateral (horizontal) semicircular canal.
13. **What is the crus commune?** The joined non-ampullated ends of the superior and posterior canals (about 4 mm).
14. **What does Rosenthal's canal contain?** The spiral ganglion, running along the osseous spiral lamina.
15. **Three walls of the cochlear duct?** Basilar membrane (floor), Reissner's membrane (roof), stria vascularis (outer wall).
16. **Connection between the saccule and cochlear duct?** The ductus reuniens.
17. **Where is the endolymphatic sac?** Between two layers of dura on the posterior surface of the petrous bone.
18. **Hair cell counts of the utricular and saccular maculae?** Utricle about 33,000; saccule about 18,000.
19. **Orientation of the utricular and saccular maculae?** Utricle horizontal; saccule vertical.
20. **What lies on the otolithic membrane?** Otoconia: calcium carbonate crystals.

21. **Ewald's law for the lateral canal?** Ampullopetal flow (toward the utricle) is excitatory.
22. **Resting discharge of vestibular neurons?** About 90 spikes per second.
23. **Where is the auditory cortex?** Superior temporal gyrus, Brodmann area 41.
24. **Which tract mediates the vestibulo-ocular reflex?** The medial longitudinal fasciculus.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.